



Exam Ace Champs

Where Champions Are Made

Quick Revision

Chapterwise

Mind-Maps

PHYSICS

**class
11**

Covers

all chapters of NCERT

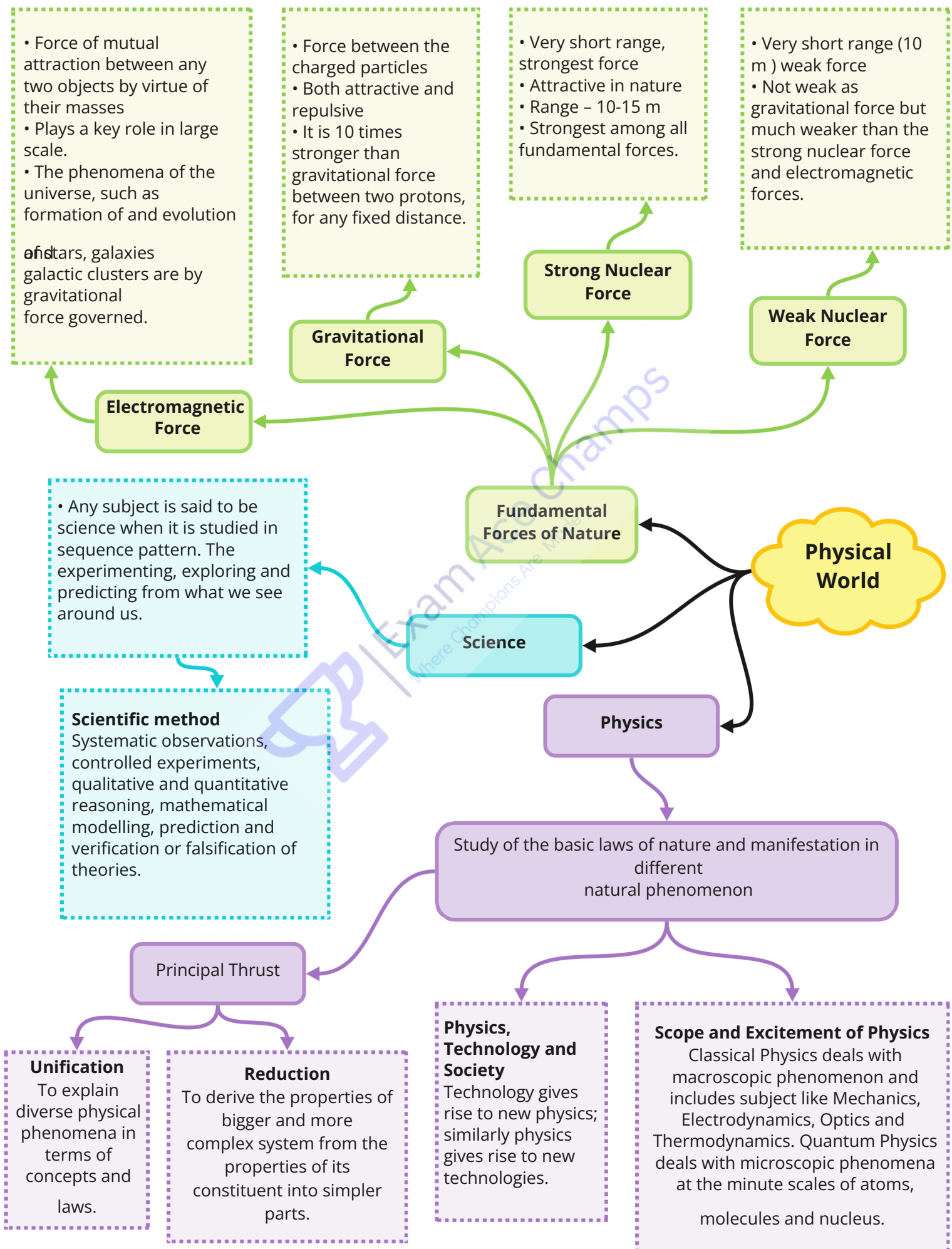


Faster Recall



Quick Revision

Class : 11th Physics
Chapter- 1 : Physical World



Class : 11th Physics

Chapter- 2 : Units and Measurements

System of units

Set of fundamental and derived units

CGS – Centimeter, Gram, Second
FPS – Foot, Pound, Second
MKS– Metre, Kilogram, Second S.I
- System International
(Internationally accepted)

Properties

Universal acceptance Non
perishable
Well defined
Does not change with time

Derived Units

* The units which are derived from fundamental units are called derived units.
* Physical quantities like speed acceleration have derived units like m/s,

$$\text{m/s}^2$$

Fundamental Units

Mass, length, time, temperature, electric current, luminous intensity and amount of substance have fundamental units. Kilogram, metre, second, Kelvin, ampere, candela and mol respectively.

Supplementary Units

Quantities plane angle and solid angle have radian and steradian as units

* Comparison with a certain internationally accepted reference standard is called unit
* Reference standard use to measure physical quantities.

Types

Multiplication or Division

$$\text{If } Z = A/B \\ \frac{\Delta Z}{Z} = \left(\frac{\Delta A}{A}\right) + \left(\frac{\Delta B}{B}\right)$$

Sum of Difference

$$\pm \Delta Z = \pm \Delta A \pm \Delta B$$

Measure Quantity raised to the power

$$\text{If } Z = A^p \times B^q / C^r \\ \frac{\Delta Z}{Z} = p\left(\frac{\Delta A}{A}\right) + q\left(\frac{\Delta B}{B}\right) + r\left(\frac{\Delta C}{C}\right)$$

Combination of Errors in

Unit

Units and Measurements

$$\text{Error} = \text{True Value} - \text{Measured Value}$$

Types

Measurement

Significant Figures

All non- zero digits
All zero between two non- zero digits
If the number is less than 1, the zero(s) on the right of decimal point but to the left of the non- digit zero are not significant.
Terminal or trailing zero (s) in a number without a decimal point are not significant
e.g In 020342.010
Here 20342.01 are significant

Dimensions of Physical Quantities

Expression which shows how and which of the base quantities represent the dimensions of a physical quantity is called dimension formula.
e.g:- Force =

$$[MLT^{-2}]$$

Precision

Limit of resolution of quantity measured.

Accuracy

Closeness of the measured value to the true value

Absolute Error

$$\Delta a_n = a_n - a_{\text{mean}}$$

$$a_{\text{mean}} = \frac{a_1 + a_2 + \dots + a_n}{n}$$

Relative Error

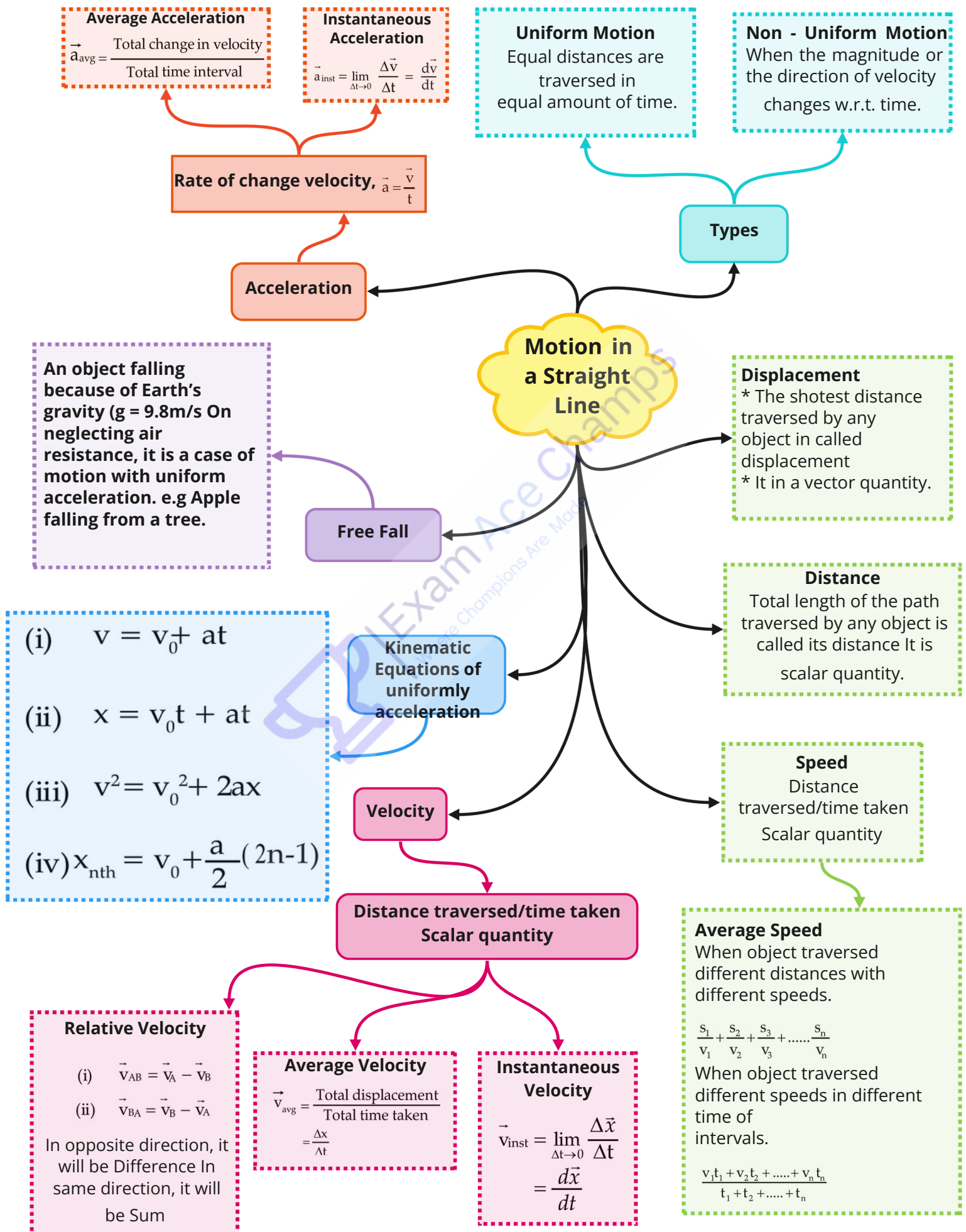
$$= \frac{\Delta a_{\text{mean}}}{a_{\text{mean}}}$$

$$\Delta a_{\text{mean}} = \frac{(|\Delta a_1| + |\Delta a_2| + \dots + |\Delta a_n|)}{n}$$

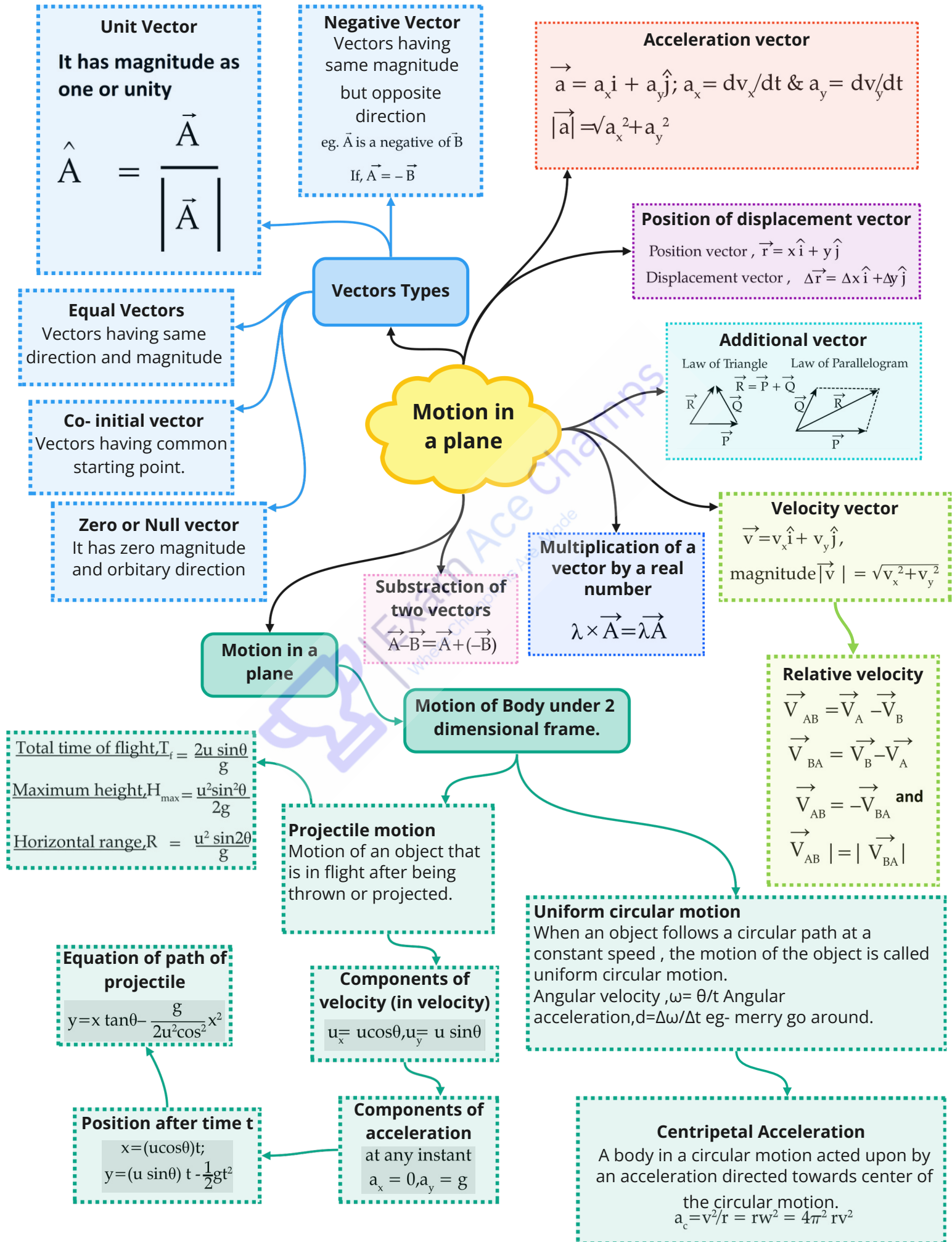
Percentage Error

$$\delta a = \frac{\Delta a_{\text{mean}}}{a_{\text{mean}}} \times 100$$

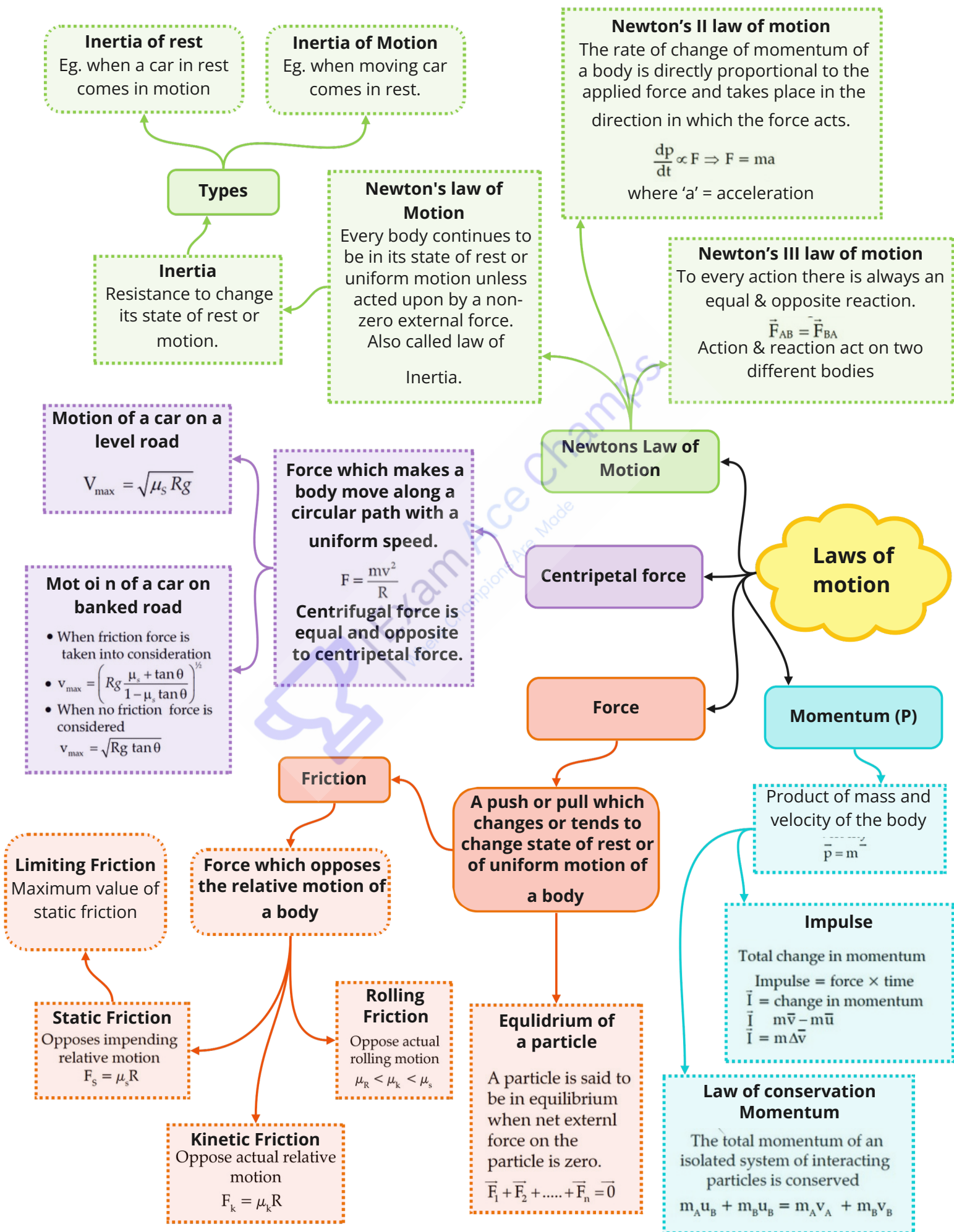
Class : 11th Physics
Chapter- 3 : Motion in a Straight Line



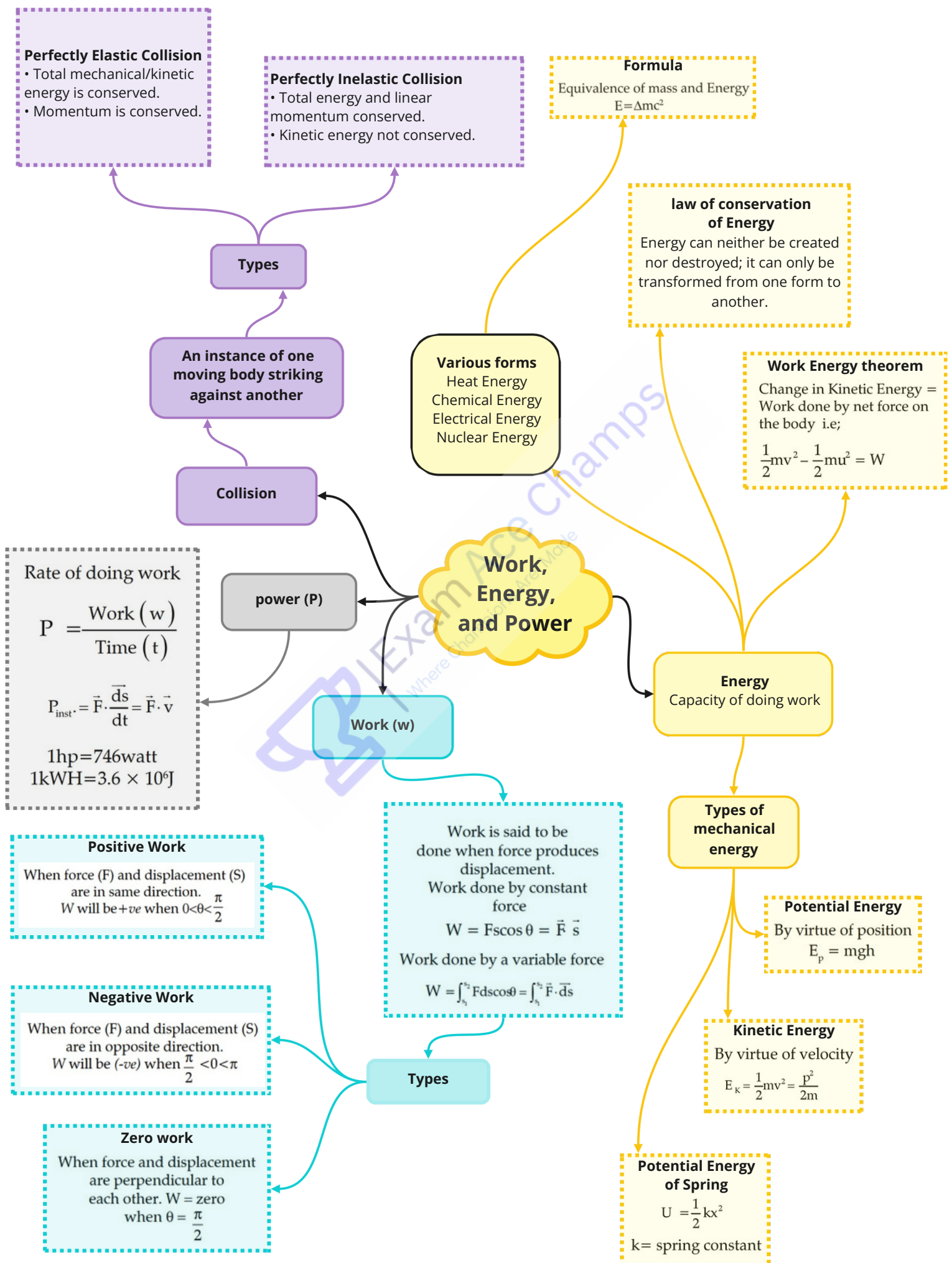
Class : 11th Physics
Chapter- 4 : Motion in a plane



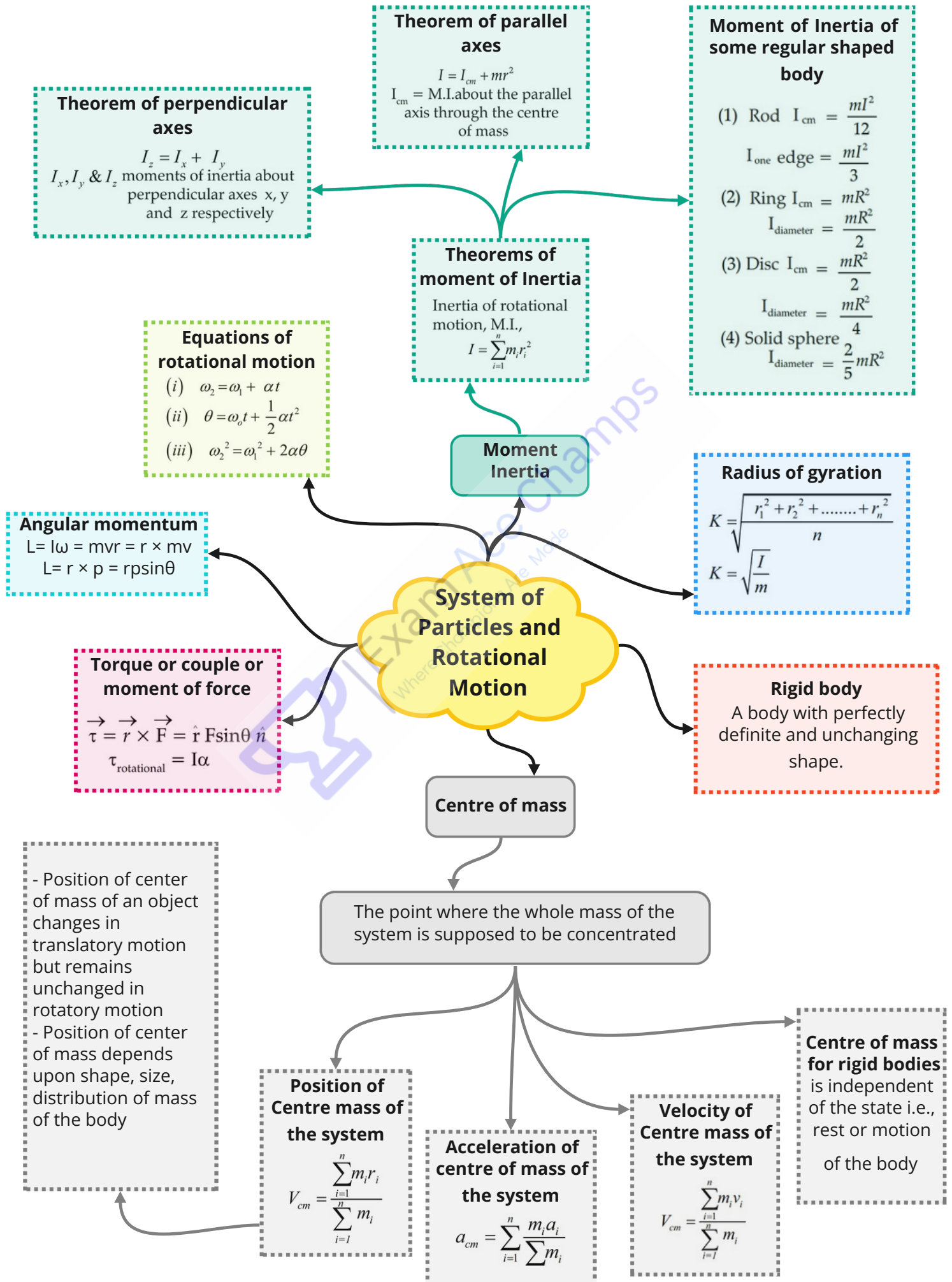
Class : 11th Physics
Chapter- 5 : Laws of motion



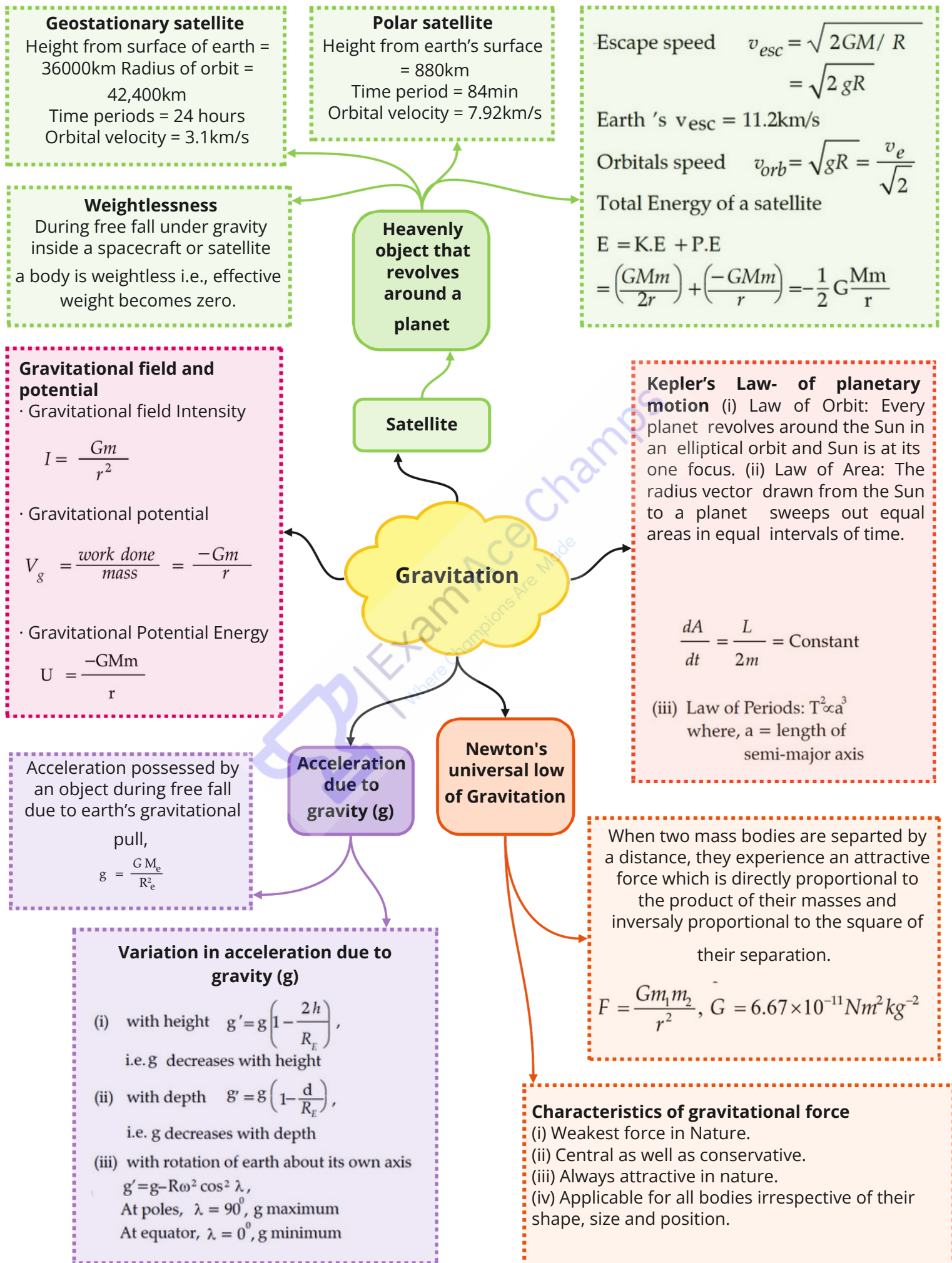
Class : 11th Physics
Chapter- 6 : Work, Energy, and Power



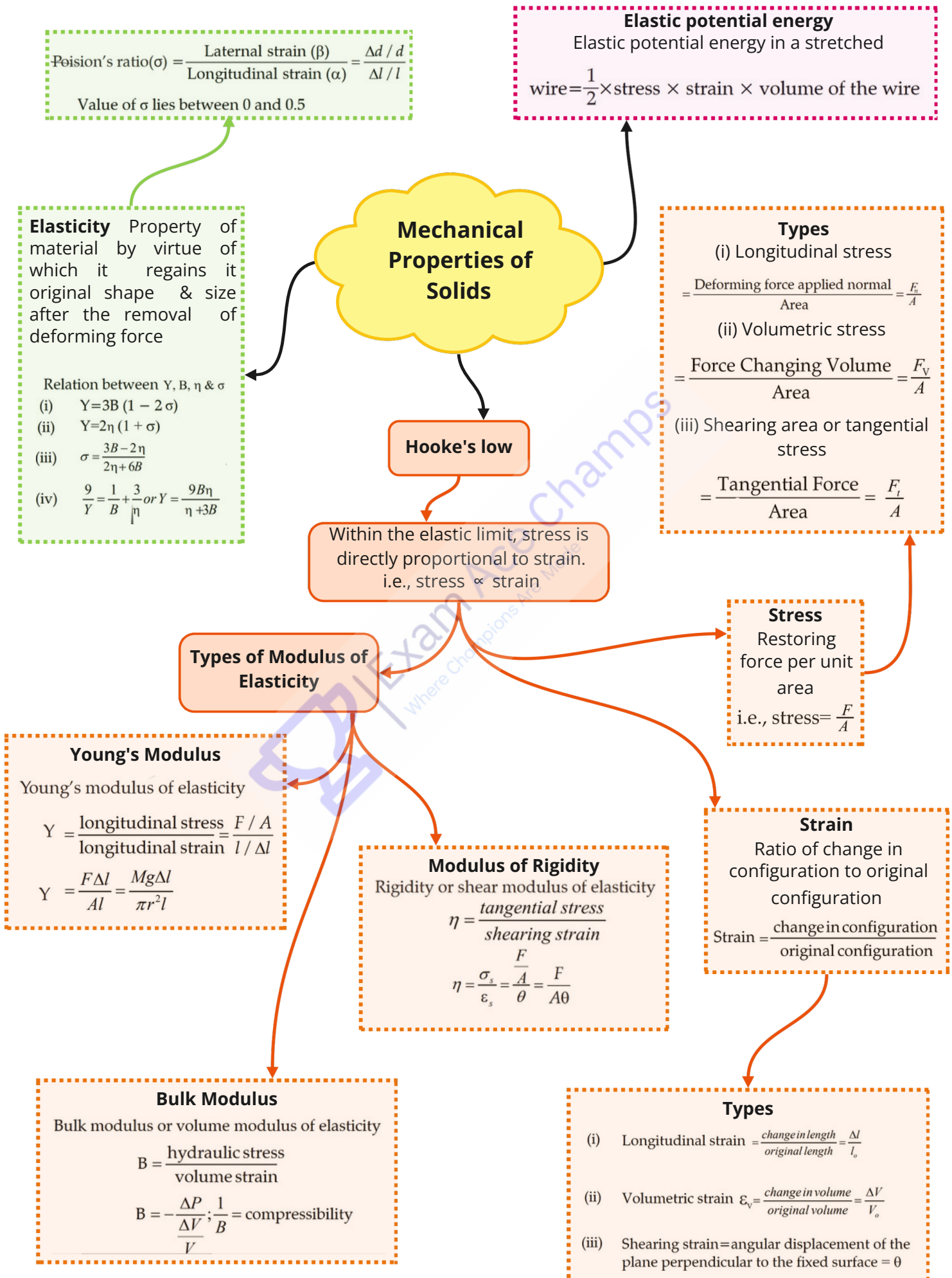
Class : 11th Physics
Chapter- 7 : System of Particles and Rotational Motion



Class : 11th Physics
Chapter- 8 : Gravitation



Class : 11th Physics
Chapter- 9 : Mechanical Properties of Solids



Class : 11th Physics
Chapter- 10 : Mechanical Properties of Fluids

Venturimeter

Device used to measure the rate of flow of liquid. Volume of liquid flowing per second.

$$Q = a_1 a_2 \sqrt{\frac{2h\rho_m g}{\rho(a_1^2 - a_2^2)}}$$

Toricell's Low

Velocity of efflux of liquid through an orifice

$$V = \sqrt{2gh}$$

Applications

- o Lift of an aircraft wing.
- o Sprayer or atomizer
- o Blowing off the roofs during windstorm.

Stroke's law $F = 6 \eta v r$

Opposing force between different layers of fluid in relative motion

Viscous drag $F = -\eta A \frac{dv}{dx}$
 η = coefficient of viscosity

Streamline : In liquid flow when the Velocity is less than critical velocity, each particle of the liquid passing through a point travels along the same path and same velocity as the preceding particles.

Turbulent : When velocity of liquid flow is greater than critical velocity and particles follow zig-zag path.

For an incompressible non- viscous streamline, irrotational flow of fluid,

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$$

Bernoulli's principal

Viscosity

Flow of fluids

Formulae

Mechanical Properties of Fluids

Laws of Fluids

That can flow like liquids and gases

Equation of continuity

$m = a_1 v_1 \rho_1 = a_2 v_2 \rho_2$
 for an incompressible liquid,
 $\rho_1 = \rho_2$ then $a_1 v_1 = a_2 v_2$
 or $av = \text{constant}$

Pascal's law : The pressure exerted at any point on an enclosed liquid is transmitted equally in all direction.

- Surface tension $S = \frac{F}{l}$
- Surface Energy = $\frac{\text{work done in increasing area}}{\text{increase in surface area}} = \frac{W}{\Delta A}$
- Capillary rise or fall, $h = \frac{2S \cos \theta}{r \rho g}$
- Excess Pressure inside a drop (liquid)
 $P_{\text{excess}} = \frac{2S}{R}$
- Excess Pressure inside a bubble (soap)
 $P_{\text{excess}} = \frac{4S}{R}$

Pressure(P) = $\frac{\text{thrust}(F)}{\text{area}(A)} = \lim_{\Delta A \rightarrow 0} \frac{\Delta F}{\Delta A} = \frac{dF}{dA}$
 Pressure exerted by a liquid column of height h, $(p) = h\rho g$

Atmospheric Pressure(Pa)

Pressure (atm) exerted by the atmosphere. At sea level, 1 atm = pressure exerted by 0.76m of Hg = $h\rho g = 0.76 \times 13.6 \times 10^3 \times 9.8 = 1.013 \times 10^5 \text{ Nm}^2 = 101.3 \text{ kPa}$

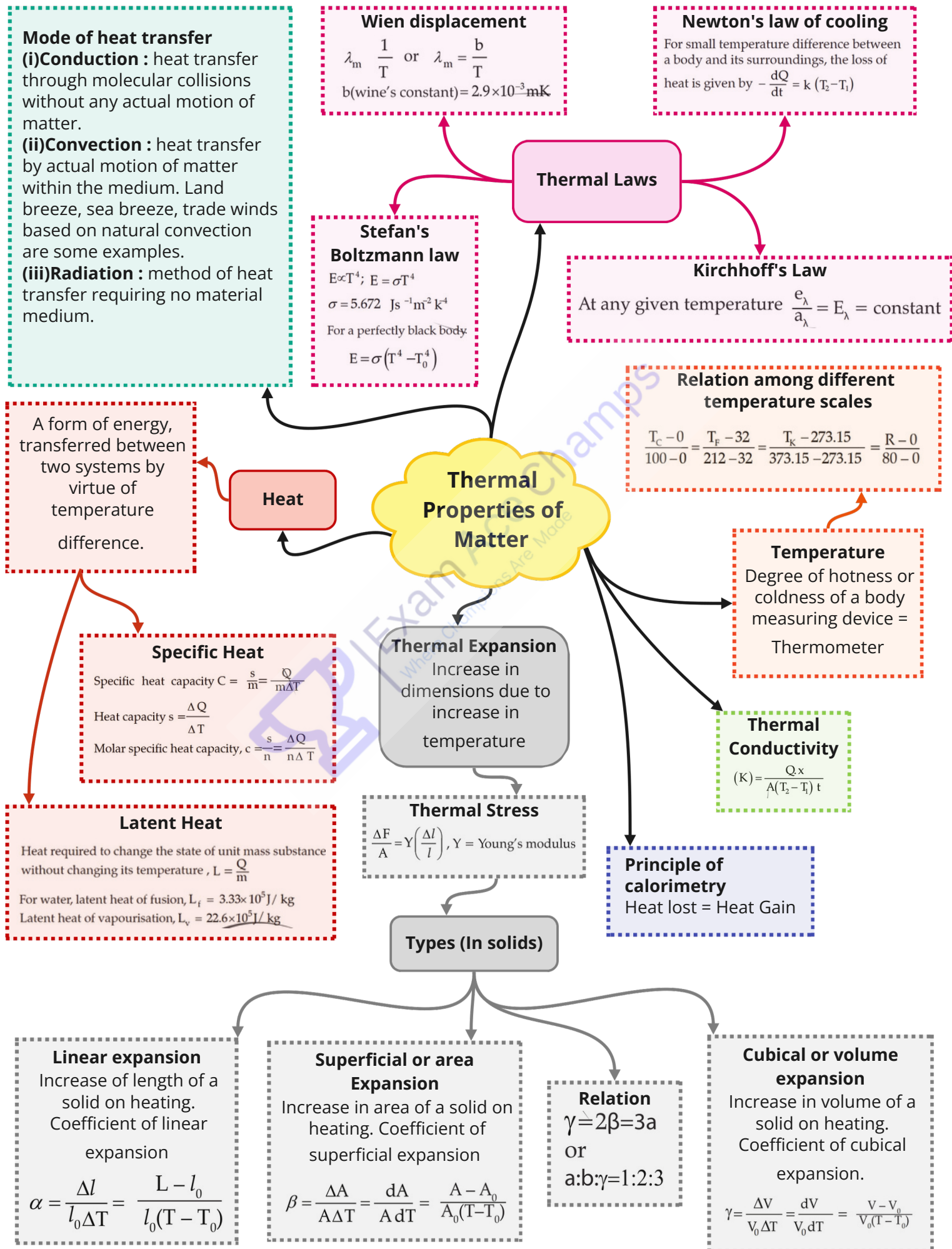
Absolute Pressure(P)

Total or actual pressure at a point. Absolute pressure = atmospheric pressure + gauge pressure

Gauge Pressure(Pg)

Difference between the absolute pressure at a point and the atmospheric pressure.
 $P_g = \text{absolute pressure (P)} - \text{atmospheric pressure (Pa)}$

Class : 11th Physics
Chapter- 11 : Thermal Properties of Matter



Class : 11th Physics
Chapter- 12 : Thermodynamics

Thermodynamics State variable

- (i) Extensive – indicate the size of the system. e.g. U, volume, total mass.
(ii) Intensive – do not indicate size of the system. e.g. , pressure & temperature

Thermodynamics Equilibrium

Two systems are in thermal equilibrium with each other if they have the same temperature.

Zeroth Low of Thermodynamics

If two systems A&B are in thermal equilibrium with a third system C, then A&B are in thermal equilibrium with each other.

First Low of Thermodynamics

It is the statement of law of conservation of Energy;
 $\Delta Q = \Delta U + \Delta W$
 $= \Delta U + P\Delta U$ (Here, $\Delta U = P\Delta U$)

Second Low of Thermodynamics

It is impossible for an engine working between a cyclic process to extract heat from a reservoir and convert completely into work.

Laws

Reversible & Irreversible Process

Any process made to proceed in the reverse direction by changing its conditions is called Reversible Process.

Any process which cannot be retraced in the reverse direction exactly is called Irreversible Process.

Heat Engine

Converts continuously heat into mechanical energy in cyclic process.

$$\text{Efficiency } \eta = \frac{W}{Q} = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

Carnot Engine

An Ideal engine works on a reversible cycle of four operations in succession.

- (i) isothermal expansion,
(ii) adiabatic expansion,
(iii) isothermal compression,
(iv) adiabatic compression.
Efficiency of Carnot's engine

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

Refrigerator and heat pump

Refrigerator is a heat engine working in the reverse direction.

Coefficient of performance,

$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

Quasi - static Process

Infinity slow process such that systems remain in thermal & mechanical equilibrium with the surroundings throughout.

Property

The pressure (P) & temperature (T) of the environment can differ from those of the system only infinitesimally.

Adiabatic Process

A thermally insulated system neither gains nor loses heat

$$PV^\gamma = \text{Constant}$$

$$TV^{\gamma-1} = \text{Constant}$$

$$\frac{P^{1-\gamma}}{T^\gamma} = \text{Constant}; \quad \frac{C_p}{C_v} = \gamma$$

Isochoric Process
Volume = Constant

$$\frac{P}{T} = \text{Constant}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Isobaric Process

Pressure = Constant

$$\frac{V}{T} = \text{Constant}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Isothermal Process

Temperature = Constant
 $PV = \text{Constant} = nRT$

$$W = \int_{V_1}^{V_2} \frac{nRT}{V} dv$$

Thermodynamics

Specific heat capacity Relation

- $\frac{C_p}{C_v} = \gamma$
- Mayer's Equation
 $C_p - C_v = R$

Branch of Science which deals with concepts of heat & temperature and their interconversion by thermodynamic process

Cyclic Process

In this process, system returns to initial state for a cyclic process $\Delta U = 0$ (zero)

Degree of freedom

For monoatomic gas: $f = 3$ For diatomic gas:

- (i) at room temperature, $f = 5$
- (ii) at high temperature, $f = 7$

For polyatomic gas:

- (i) at room temperature, $f = 6$
- (ii) at high temperature, $f = 8$

Assumption of kinetic Theory of Gases

- All the molecules of a gas are identical.
- The molecules of different gases are different.
- The molecules of gases are in a state of random motion.
- The collisions of gases molecules are perfectly elastic.

Average Speed

$$v_{av} = \frac{v_1 + v_2 + v_3 + \dots + v_N}{N}$$

$$v_{av} = \sqrt{\frac{8RT}{\pi M}} = \left(\sqrt{\frac{8}{3\pi}}\right) v_{rms} = 0.92 v_{rms}$$

Pressure exerted by gas

$$P = \frac{1}{3} \rho v_{rms}^2$$

Mean Free Path

$$\lambda = \frac{1}{\sqrt{2} n d^2} = \frac{k_B T}{\sqrt{2} \pi d^2 P}$$

Root mean square speed

$$v_{rms} = \sqrt{\frac{3PV}{M}} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M}}$$

Most probable speed

$$v_{mp} = \sqrt{\frac{2RT}{M}} = \left(\sqrt{\frac{2}{3}}\right) v_{rms} = 0.816 v_{rms}$$

Kinetic Theory

Relation between pressure (p) and Kinetic Energy (E)

$$E = \frac{3}{2} PV$$

Behaviour of Gases

Law of Equipartition Energy

Energy associated with each degree of freedom per molecule = $\frac{1}{2} kT$

Specific Heat Capacity

Specific Heat Capacity for an ideal gas, $C_p - C_v = R$

for monoatomic $\frac{C_p}{C_v} = \gamma = \frac{5}{3}$

for diatomic $\frac{C_p}{C_v} = \gamma = \frac{7}{5}$

for polyatomic gases, $\frac{C_p}{C_v} = \gamma = \frac{(4+f)}{(3+f)}$

Hence, f is the degree of freedom

An ideal gas satisfies equation $PV = nRT$
all pressure and temperature $n =$ no. of moles, $R = N_A k_B$ universal gas constant

Charles's Law

$V \propto T$ (If $P =$ constant)

$$\frac{V}{T} = \text{constant}$$

$$\text{or } \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Avogadro's law

Under the same condition of temperature and pressure equal volumes of all gases contain equal no. of molecules.

$$\text{i.e. } N_1 = N_2$$

Boyles's Law

$P \propto \frac{1}{V}$ (If $T =$ constant)

$$PV = \text{constant}$$

$$\text{or } P_1 V_1 = P_2 V_2$$

Guy Lussac's Law

$P \propto T$ (If $V =$ constant)

$$\frac{P}{T} = \text{constant}$$

$$\text{or } \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Dalton's Law of partial pressure

Total pressure of a mixture of non - reacting gases,
 $P = P_1 + P_2 + \dots + P_n$

Class : 11th Physics
Chapter- 14 : Oscillations

Velocity in SHM

$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$$

$$v_{\max} = \omega A$$

Acceleration in SHM

$$a = \frac{dv}{dt} = -\omega^2 A \cos(\omega t + \phi)$$

$$a_{\max} = \omega^2 A$$

Energy in SHM

Kinetic Energy

$$K.E = \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t + \phi)$$

Potential Energy

$$P.E = \frac{1}{2} m \omega^2 A^2 \cos^2(\omega t + \phi)$$

Total energy, T.E. = K.E. + P.E.

$$= \frac{1}{2} k A^2 [\cos^2(\omega t + \phi) + \sin^2(\omega t + \phi)]$$

Displacement in SHM

$$x(t) = A \cos \omega t + \phi$$

A = Amplitude i.e., maximum displacement of particles.

Phase

Time varying quantity $(\omega t + \phi)$

Phase constant or Phase angle (ϕ)

- It depends upon velocity (v) and displacement of particle at $t = 0$

Simple Harmonic Motion is the simplest form of oscillatory motion.

SHM

Force law equation for SHM

$$F = -k(x), k = m\omega^2$$

$$\therefore \omega = \sqrt{\frac{k}{m}}$$

Time period(T).

The smallest interval of time after which the motion is repeated

Frequency

The number of oscillations per second.

$$\text{i.e., } f = \frac{1}{T} = \frac{\omega}{2\pi}$$

Oscillations

Oscillations due to a spring

$$\text{Time period } T = 2\pi \sqrt{\frac{m}{k}}$$

$$\text{Here, } k = \frac{F}{x} = \text{spring constant}$$

Simple pendulum

$$\text{Time period } T = 2\pi \sqrt{\frac{l}{mg}} = 2\pi \sqrt{\frac{l}{g}}$$

- Every oscillatory motion is periodic, but every periodic motion need not be oscillatory.
- To and from motion repeatedly about a fixed point in a definite interval of time.

Damped Oscillations

Oscillation of a body whose amplitude goes on decreasing with time.

Damping force $F_d = -bv$, b = damping constant

Displacement of damped oscillator

$$x = x_m e^{-bt/2m} \sin(\omega t + \phi)$$

Amplitude decreases with time as

$$x = x_m e^{-(b/2m)t}$$

Forced Oscillations

Driving force, $F(t) = F_0 \cos \omega_d t$

Displacement, $x(t) = A \cos(\omega_d t + \phi)$

$$\text{Amplitude, } A = \frac{F_0}{\{m^2(\omega^2 - \omega_d^2)^2 + \omega_d^2 b^2\}^{1/2}}$$

$$\tan \phi = \frac{-v_0}{\omega_d x_0}$$

Resonance

When frequency of external force is equal to natural frequency of oscillator.

**Relation b/w wave velocity (v)
Wavelength (y) & frequency (v)**

$$v = v\lambda$$

$$\text{Time period (T)} = \frac{1}{\text{Frequency}} = \frac{1}{v}$$

$$\text{Angular Frequency } (\omega) = 2\pi v$$

$$\text{Wave number, } v = \frac{1}{\text{wavelength}} = \frac{1}{\lambda}$$

Beats

Difference in frequencies of two superposing waves,

$$v_{\text{beat}} = v_1 \sim v_2$$

Displacement relation in a plane progressive wave

$$y(x, t) = A \sin(-kx + t)\omega_\phi$$

$$y(x, t) = A \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

Velocity amplitude $v_0 = A\omega^2$
Acceleration Amplitude $a_0 = A\omega^2$

Principle of superposition of waves

When two or more waves are propagating with different displacements then the net displacement of collective wave is given as $Y = y_1 + y_2 + y_3 + \dots + y_n$

Waves

Doppler Effect

General Formula, Apparent Frequency

$$n' = n \left(\frac{v \pm v_s \pm v_m}{v \mp v_0 \pm v_m} \right)$$

Here, v = speed of sound,
 v_0 = observer speed
 v_s = source speed
 v_m = medium speed

Stationary Waves

Stationary waves

(i) In string: Fundamental Frequency, $v = \frac{1}{2l} \sqrt{\frac{T}{m}}$

(ii) In organ pipe:
(a) Open at both ends,

Fundamental freq. or 1 harmonic $v = \frac{v}{2l}$

(b) Closed at one end $v = \frac{v}{4l}$

(iii) • In open organ pipe, higher harmonics, both odd & even.
• In closed organ pipe, higher harmonics, odd only.

Types

Matter Waves

Waves associated with particles like- electrons, protons, neutrons, atoms, molecules etc.

Mechanical Waves

Require material medium for their propagation

Electromagnetic Waves

Do not require any material medium for their propagation e.g., light waves

Transverse Waves

Individual particles of the medium oscillate perpendicular to the direction of propagation of wave

Longitudinal Waves

Individual particles of the medium oscillate along the direction of propagation of wave.

Properties

Essential properties for propagation
- Elasticity
- Inertia
- Minimum friction

Speed of transverse wave in a stretched string

$$V = \sqrt{\frac{T}{m}}, T = \text{tension} \ \& \ M = \text{mass/length}$$

$$\text{Speed of a longitudinal wave speed of sound, } V = \sqrt{\frac{B}{\rho}}$$

$$= \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{P}{\rho}} = \sqrt{\frac{\gamma P}{\rho}} \text{ for air, } \gamma = 7/5$$



India's favourite IIT-JEE & NEET Study Bundle

- ✓ Toppers Handwritten Notes
- ✓ Previous year papers with Solution
- ✓ Test Series & Mock Tests
- ✓ Formula Book - Chapter wise
- ✓ Short Notes for Quick Revise
- ✓ Mind Maps - for easy visualization
- ✓ Question Bank and more

Ready to Crush IITJEE or NEET?
Everything you need. One powerful pack.
✦✦ Study smarter. Rank higher.

Click here



Don't wait! Visit **examacechamps.com** and grab your Success Bundle now.